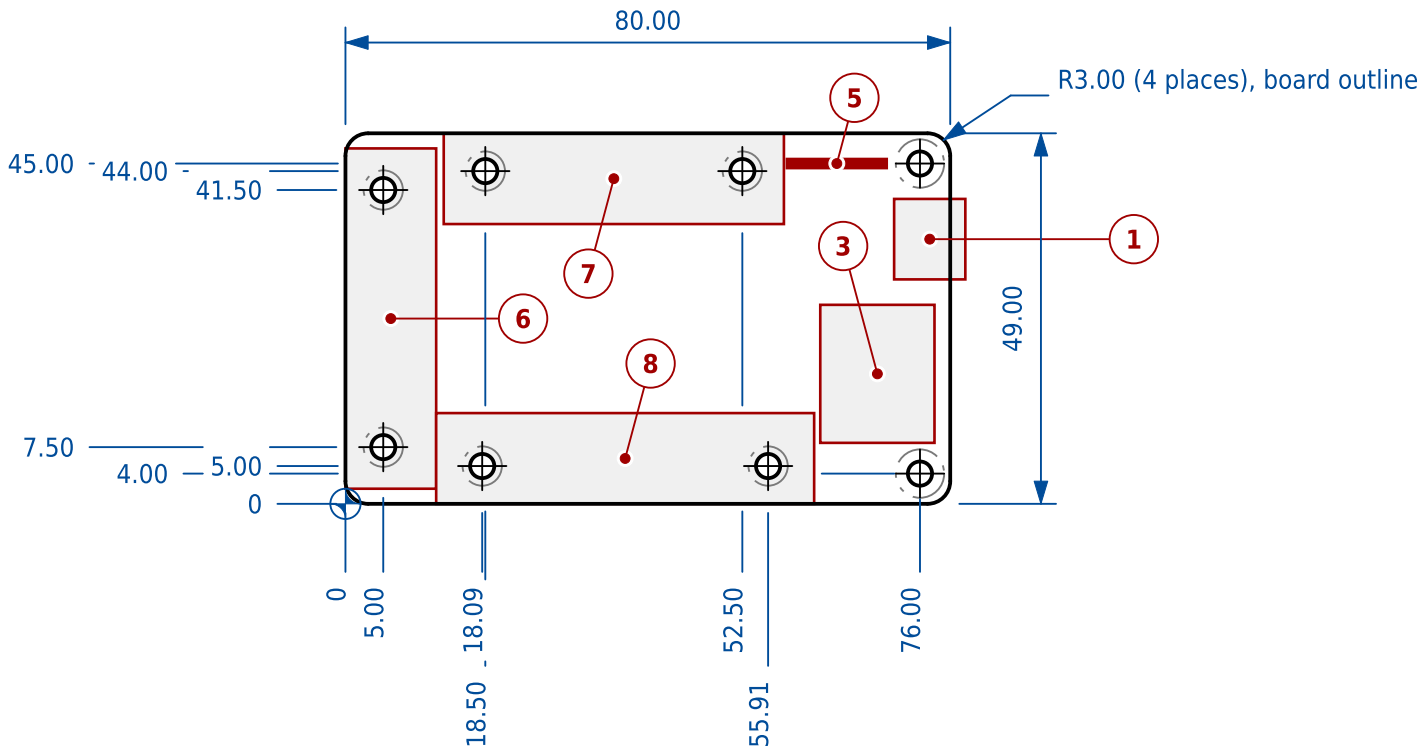




HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H1	76.00	45.00	3.20	6.40
H2	76.00	4.00	3.20	6.40
SA1	5.00	7.50	3.20	5.20
SA2	5.00	41.50	3.20	5.20
SB1	18.50	44.00	3.20	5.20
SB2	52.50	44.00	3.20	5.20
SC1	18.09	5.00	3.20	5.20
SC2	55.91	5.00	3.20	5.20

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C J1, programming and console	72.58 to 82.00	29.68 to 40.32
2	Second USB port - not on this board	-	-
3	Ethernet RJ45 CON1	62.81 to 77.92	8.06 to 26.32
4	User LEDs, single colour - not on this board	-	-
5	RGB LEDs D1-D7, 7 on 2.00 mm pitch	58.40 to 71.60	44.40 to 45.60
6	SYZYGY port A, J3	0.00 to 12.00	2.00 to 47.00
7	SYZYGY port B, J4	13.00 to 58.00	37.00 to 49.00
8	SYZYGY port C, J5	12.00 to 62.00	0.00 to 12.00
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs H1 and H2 are the KiCad reference designators; SA1 to SC2 are the SYZYGY standoff holes of ports A, B and C, plated, and the maker's acrylic plate bolts through all eight.
- No Pmod host: expansion is three SYZYGY ports.
- Assembled envelope, connector overhang included: 82.00 x 49.00 mm.

SOURCES

- KiCad board file:
https://github.com/butterstick-fpga/butterstick-hardware/blob/main/ButterStick_r1.0/ButterStick.kicad_pcb @ fac6a834 - title block: ButterStick rev r1.0a, dated 2021-01-10; Greg Davill / GsD
- Board size:
<https://github.com/butterstick-fpga/butterstick-hardware/blob/main/README.md> - quoted: "Board dimensions: 80mm x 49mm"
- Thickness:
https://github.com/butterstick-fpga/butterstick-hardware/blob/main/hardware/ButterStick_r1.0/Production/ButterStick-r1.0a-fab-notes.txt - quoted: "Finished board thickness is 1.6 mm."

BUTTERSTICK

r1.0			
DRAWN		VERSION	UNITS
generated		v0.0-270-gc2d9e2f	mm
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	FPGA-BUTTERSTICK	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	